Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)
Physical DBMS	Indexing	Which of the following is true regarding indexing in a physical database management system? A) Indexing is used to speed up the search for data within a table. B) Indexing always results in faster query processing time. C) Indexing is only used for primary keys. D) Indexing is not necessary if a table is small. Correct answer: A) Indexing is used to speed up the search for data within a table.
Super key	Number of super keys possible	If R(A1 A2 A3......A100000) then how many super keys are possible? a) If A1 is candidate key b) If A1 and A2 is candidate key c) If A1 A2 and A3 A4 is candidate key Ans- a) 2^(n-1) = 2^(100000-1)= 2^99999 b) 2^(n-1)+ 2^ (n-1) - 2^(n-2) = 2^100000-2^99998= c) 2^(n-2) + 2^(n-2) -2^(n-4) = 2^99999-2^99996
Transactions	ACID Properties of Transactions	If a transaction fails after deducting amount from account A and before crediting the amount in account B leading to money loss, what type of requirement does this database fails? (a) Atomicity (b) Durability (c) Consistency (d) Isolation Answer is (a) Atomicity
Physical storage	Storage Heirarchy	which one of the following is Primary Storage? a. Cache b. flash memory c. magnetic disks d. optical storage Answer a. Cache

OLAP Queries	SQL aggregation functions and GROUP	Which of the following SQL commands will correctly return the average salary of employees in each department in a table named "employees"?
		a. SELECT department, AVG(salary) FROM employees;
		b. SELECT AVG(salary) FROM employees GROUP BY department;
		c. SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department;
		d. SELECT AVG(salary) AS avg_salary, department FROM employees;
		Answer: c. Option c correctly groups the results by department, calculates the average salary for each department, and assigns the result to an alias named "avg_salary".

Transactions	ACID Properties	which of the following database component who cares about consistency: a) Concurrency control component b) Transaction management component c) Recovery management component d) individually none of the above e) if these three happens together then option a,b,c all correct. answer is d) and e)

Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)
Transaction	Locks in Transaction	Suppose that a DBMS recognizes increment, which increments an integer-valued object by 1, and decrement as actions, in addition to reads and writes. A transaction that increments an object need not know the value of the object; increment and decrement are versions of blind writes. In addition to shared and exclusive locks, two special locks are supported: An object must be locked in I mode before incrementing it and locked in D mode before decrementing it. An I lock is compatible with another I or D lock on the same object, but not with S and X locks. 1. Illustrate how the use of I and D locks can increase concurrency. (Show a schedule allowed by Strict 2PL that only uses S and X locks. Explain how the use of I and D locks can allow more actions to be interleaved, while continuing to follow Strict 2PL.) 2. Informally explain how Strict 2PL guarantees serializability even in the presence of I and D locks. (Identify which pairs of actions conflict, in the sense that their relative order can affect the result, and show that the use of S, X, I, and D locks according to Strict 2PL orders all conflicting pairs of actions to be the same as the order in some serial schedule.) Answer : 1. Take the following two transactions as example: T1: Increment A, Decrement B, Read C; T2: Increment B, Decrement A, Read C If using only strict 2PL, all actions are versions of blind writes, they have to obtain exclusive locks on objects. Following strict 2PL, T1 gets an exclusive lock on A, if T2 now gets an exclusive lock on B, there will be a deadlock. Even if T1 is fast enough to have grabbed an exclusive lock on B first, T2 will now be blocked until T1 finishes. This has little concurrency. If I and D locks are used, since I and D are compatible, T1 obtains an I-Lock on A, and a D-Lock on B; T2 can still obtain an I-Lock on B, a D-Lock on A; both transactions can be interleaved to allow maximum concurrency. 2. The pairs of actions which conflicts are: RW, WW, WR, IR, IW, DR, DW We know that strict 2PL orders the first 3 conflicts pairs of actions to be the same as the order in some serial schedule. We can also show that even in the presence of I and D locks, strict 2PL also orders the latter 4 pairs of actions to be the same as the order in some serial schedule. Think of an I (or D)lock under these circumstances as an exclusive lock, since an I(D) lock is not compatible with S and X locks anyway (ie. can't get a S or X lock if another transaction has an I or D lock). So serializability is guaranteed.
Schedules	Serializable Schedules	SHORT ANSWER Q.When are 2 schedules said to be equivalent? A.If for any database state, the effect of executing the 1st schedule is identical to the effect of executing the 2nd schedule then they are said to be equivalent.
B+ Trees	Updates on B+ Trees	Question. State whether the following statements are True or False: In the case of data insertion in a B+ Tree, A. Every data entry must appear only once in a leaf node. B. Splitting is usually better than redistribution in non-leaf nodes. C. Checking for possibility of redistribution in leaf nodes increases I/O. D. On splitting index entries, the middle key is "pushed up". E. On splitting leaf nodes, previous and next neighbor pointers must be updated. Answer. A. False B. True C. False D. True E. True Reference: Lecture slides Week 8, B+ Trees and hashing
Transaction	Transaction Schedules	Question: What is the advantage of a serial schedule over a parallel schedule? Answer: Serial schedules are 'consistent', while parallel schedules are generally not.
Indexing - B+ Trees	Spatial and Temporal Data	How is a BT Tree different than that of a R+ Tree? Ans: Both BT & R+ trees are special types of B+ trees, BT-trees optimise time-series queries by organising nodes based on times intervals. Whereas, R+ trees optimise spatial queries.

Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)
Transaction	ACID properties	Ques) Which is the correct statement about the ACID properties of a database transaction? Option A: If a transaction commits successfully, then all ACID properties must be true as it ensures that all writes are persisted before the transaction is completed. Option B: The ACID properties are only applicable to serializable transactions that maintains a strict ordering of operations. Option C: The ACID properties can be weakened to improve performance sometimes, however this may result in discrepancies and data corruption. Option D: It is impossible to achieve four ACID properties together as there is an inherent tradeoffs between them. Answer : Option C Justification about every option: Option A: Incorrect as ACID properties guarantees atomicity, consistency, isolation and durability but there are other factors like deadlock or violation of integrity constraints due to which Option B: Incorrect as in case of non serializable transactions as well sometimes ACID properties are preserved and even when not there are several mechanism like by using exclusive Option C: Correct: This is indeed true that sometimes weaker isolation levels could save time and increase performance of transaction but we need to handle these things with care as t Option D: Incorrect: Yes, it is very tough to achieve all properties together but it is not impossible as by using appropriate locking mechanisms, durable storage and transaction isolation
Indexing and Hashing	Indexing and Hashing	Assume you are designing a database for a large online retailer that sells millions of products. In order to efficiently retrieve data on customer orders, you decide to implement an indexi (a) Define the key attributes for the "Orders" table and explain your rationale for selecting them. (b) Create a B-tree index on the "Orders" table using the key attributes you selected in part (a). Show the resulting B-tree structure and explain how it can be used to efficiently retrieve (c) Implement a hash table for the "Orders" table using the same key attributes. Explain how you would choose the hash function and handle collisions to minimize the number of disk re (d) Compare the performance of the B-tree index and hash table for different query types (e.g. retrieving orders for a specific customer, retrieving orders within a certain date range, etc. ANSWER (a) The key attributes for the "Orders" table should be selected carefully in order to efficiently retrieve data on customer orders. A good choice for the key attributes would be a combinat (b) To create a B-tree index on the "Orders" table, we first need to choose a suitable index structure. A B-tree is a good choice for this scenario, as it allows for efficient range queries on (c) To implement a hash table for the "Orders" table using the customer ID and order date attributes, we would first choose a suitable hash function. A good choice for this scenario woul hash(key) = (a * customer_id + b * order_date) % N where a and b are random coefficients, and N is the size of the hash table. To handle collisions, we could use an open addressing scheme such as linear probing or quadratic probing, v (d) The performance of the B-tree index and hash table would depend on several factors, including the size of the "Orders" table, the distribution of data, and the specific query types be
Transactions	Deadlocks	objective-type question q. How can the concurrency controller prevent starvation of aborted transactions,while resolving deadlocks? ans. by assigning timestamps , to ensure higher timestamps are executed before transactions given a lower timestamp.
Transactions and concurrency c	Clustering	An index is clustered, if A. It is on a set of fields that form a candidate key. B. It is on a set of fields that include the primary key. C. The data records of the file are organized in the same order as the data entries of the index. D. The data records of the file are organized not in the same order as the data entries of the index. Ans : C
OLAP QUERIES	3 types of OLAP QUERIES, definition	Give 3 types of OLAP queries.
Transaction	ACID	Explain the concept of ACID properties in DBMS? Answer: ACID properties is the combination of Atomicity, Consistency, Isolation, and Durability properties. These properties are very helpful in allowing a safe and secure way of sharing Atomicity: This is based on the concept of “either all or nothing” which basically means that if any update occurs inside the database then that update should either be available to all the Consistency: This ensures that the consistency is maintained in the database before or after any transaction that takes place inside the database. Isolation: As the name itself suggests, this property states that each transaction that occurs is in isolation with others i.e. a transaction which has started but not yet completed should be Durability: This property states that the data should always be in a durable state i.e. any data which is in the committed state should be available in the same state even if any failure or

Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)
B+ Trees	B+ tree Advantages	What is the advantage of using B+ trees over other data structures in a database management system, and how does it improve query performance? B+ trees are commonly used in database management systems as an indexing technique to improve query performance. The advantages of using B+ trees over other data structures s Efficient search and retrieval: B+ trees are designed to efficiently search and retrieve data by minimizing the number of disk accesses required to find the desired record. This is achieve Ordered traversal: B+ trees are ordered data structures, which means that the keys are stored in a sorted order. This makes it easy to perform range queries and retrieve data in a sorte Support for large datasets: B+ trees are designed to handle large datasets efficiently. They are balanced trees that ensure that each leaf node contains the same amount of data, which Cache efficiency: B+ trees are designed to be cache-friendly. The leaf nodes of the tree are linked together in a linked list, which makes it easy to perform sequential access to the data. Overall, using B+ trees as an indexing technique in a database management system can greatly improve query performance by reducing the number of disk I/O operations required to r
transaction	schedule	Q two transaction t1 and t2 with 3 and 4 element respectively .find no. of schedule? ans = (3+4)!/(3!*4!)=35
ER diagram	weak entity	Given a relational schema, identify the weak entity and justify
Primary key	primary key	Q: What is a primary key in a database? A: A primary key is a column or a set of columns in a database table that uniquely identifies each row in the table. It is used to enforce the entity integrity of the table, which means that e
non serializable schedule	four ACID principles	which of the four ACID principles are violated by a non-serializable schedule? a. Atomicity b. Consistency c. Isolation d. Durability Answer : Isolation(option c)
Physical Storage Systems	Magnetic Disks	QUE}} What are Performance Measures of Disks? ANS}} 1} Access Time which consists of seek time(correct tract) and rotational latency(correct sector) 2} Data-transfer rate in which data is stored or retrieved from disc
Physical Storage	Redundant Arrays of Independent Disks	QUE}} What are Factors in choosing RAID level? ANS}} - Monetary cost - Performance: Number of I/O operations per second, and bandwidth during normal operation - Performance during failure - Performance during rebuild of failed disk (Including time taken to rebuild failed disk)
indexing and hashing	bitmap indices	explain how you would efficiently implement bitmap operations.
Indexing	Keys	A _____ key is sorted in order to make it easy to access data corresponding to it. Answer: Primary Explanation: A primary key is sorted in order to make it easy to access data corresponding to it
Deadlocks	Deadlock detection	Q) How is deadlock detection carried out in a dbms? Explain with a schedule and a corresponding graph example. Ans) Deadlocks are detected when a cycle is found in a waits-for-graph. Each transaction represents a node. There is an edge from transaction A to transaction B if A is waiting for B to (example of schedule and graph can be taken from slides)
File Organization	Variable length records	Q. Variable length attributes are represented in the records by which of the following combinations? (a) offset, memory address (b) offset, length (c) length, memory address (d) offset, length, memory address Ans. (B) offset, length
B+ Tree	Insertion in B+ tree	Q. What is the time complexity of insertion in a B+ Tree? Ans. O(log n)

Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)			
B+ Trees	Insertion of elements	Q) Difference between splitting and redistribution ? When to use either of them? -In case of leaf nodes: Previous and next-neighbor pointers must be updated upon insertions (if splitting is to be pursued) Hence, checking whether redistribution is possible does not increase I/O Therefore, if a sibling can spare an entry, re-distribute -In case of Non-leaf nodes: Checking whether redistribution is possible usually increases I/O Splitting non-leaf nodes is the way to go about it.			
Transaction	Transaction concept	Full form of ACID properties? Answer: Atomicity, Consistency, Isolation and durability			
Physical Storage	Storage Interfaces	Which among the three disk interface standard families supports data transfer rates of up to 24 gigabits/sec? A. SATA - (Serial ATA) B. SAS - (Serial Attached SCSI) C. NVMe (Non-Volatile Memory Express) More options can be given with different full forms. Ans: C			
Physical storage	Hardware Issues	Q) what are Hardware Issues. Ans) Software RAID: RAID implementations done entirely in software, with no special hardware support. Hardware RAID: RAID implementations with special hardware Use non-volatile RAM to record writes that are being executed Beware: power failure during write can result in corrupted disk			
Physical Storage	Optimization of Disk Block Access	Q1 How to do Optimization of Disk-Block Access Solution:It can be done by Buffering: in-memory buffer to cache disk blocks Read-ahead: Read extra blocks from a track in anticipation that they will be requested soon Disk-arm-scheduling algorithms re-order block requests so that disk arm movement is minimized elevator algorithm			
File Organization	Column-Oriented Storage	Que}} Benefits of Columnar Representation: Ans}} Reduced IO if only some attributes are accessed Improved CPU cache performance Improved compression Vector processing on modern CPU architectures			
One word, short answer, objecti	Query processing	Which sorting technique is used in I/O Cost analysis? Ans: Merge sort			
fill in blank, short answer, mcq	query processing	which sorting technique is used in I/O cost analysis? answer - merge sort			
Physical Storage Media	The________ is the fastest and most co	Cache Memory			
Transactions	Schedules	Q How many serial schedules are possible with n transactions Solution: If n = number of transactions, then number of serial schedules possible = n!			
transactions	transactions	Select the correct option for transactions: A) A transaction is a single, isolated unit of work that must be executed as a whole or not at all. B) Transactions can be executed partially, as long as the database remains in a consistent state. C) A transaction cannot be rolled back once it has been committed to the database. D) A transaction can be undone using undo command after it has been committed. Correct answer is (A)			
Query processing and optimizat	Equivalence rules	Question: State any three equivalence rules in query optimization with examples Answer: The three equivalence rules are: (i) $\sigma\theta1 \land \theta2 (E) \equiv \sigma\theta1 (\sigma\theta2 (E))$ (ii) $E1 \bowtie E2 \equiv E2 \bowtie E1$ (iii) $(E1 \bowtie E2) \bowtie E3 \equiv E1 \bowtie (E2 \bowtie E3)$			

Topic of the question	Subtopic of the question	Your Question along with the answer (either objective type or short answer type)			
Indexing	Indexing	Which of the following statements best describes indexing in DBMS? A) Indexing is a technique used to remove duplicate data from a database. B) Indexing is a method used to secure sensitive data in a database. C) Indexing is a technique used to improve query performance in a database by creating a data structure that maps values in a column to their physical locations. D) Indexing is a technique used to encrypt data in a database to protect it from unauthorized access. Answer: C) Indexing is a technique used to improve query performance in a database by creating a data structure that maps values in a column(s) to their physical locations.			
Query Processing	How does clustering of index compariso	A clustering ordered index (eg : B+ tree index) can be done in 2 ways based on the method of comparison 1st for (A >= V), using index to find first index entry ≥ v and scan index sequentially from there, to find pointers to records. 2nd for (A< V), scan leaf pages of index finding pointers to records, till first entry > v			
Transactions	Problems Due to Concurrent Schedules	Ques Identify the problems in the given Schedule: S : R2(A),R2(B), R1(A), R1(B), W1(A), R2(A) Solution : Dirty Read Problem: due to W1(A) and R2(A) Unrepeatable Read Problem: Occur because the value read by first R2(A) will get changed by W1(A)			
Query Processing	How does increasing the number of buff	Increasing the number of buffer pages in B-Way Merge Sort minimizes the number of passes and accordingly reduces the I/O cost. Specifically, the I/O cost is equal to 2N multiplied by			